TESTBENCH.SV kodu

```
module tb_mux4to1_8bit;
   logic [7:0] in0, in1, in2, in3;
   logic [1:0] sel;
   logic [7:0] out;
   mux4to1_8bit dut (in0, in1, in2, in3, sel, out);
   initial begin
      $dumpfile("dump.vcd");
      $dumpvars(0, tb_mux4to1_8bit);
      in0 = 8'hAA; in1 = 8'hBB; in2 = 8'hCC; in3 = 8'hDD;
      $display("--- 8-Bit 4:1 MUX Testi Basliyor ---");
      $display("Girisler: in0=%h, in1=%h, in2=%h, in3=%h", in0, in1, in2, in3);
      sel = 0; #10;
      $display("Zaman = %0t | sel = %b | Cikis = %h (Beklenen: AA)", $time, sel, out);
      sel = 1; #10;
      $display("Zaman = %0t | sel = %b | Cikis = %h (Beklenen: BB)", $time, sel, out);
      sel = 2; #10;
      $display("Zaman = %0t | sel = %b | Cikis = %h (Beklenen: CC)", $time, sel, out);
      sel = 3; #10;
      $display("Zaman = %0t | sel = %b | Cikis = %h (Beklenen: DD)", $time, sel, out);
      $display("--- Test Tamamlandi ---");
      $finish;
   end
endmodule
```

DESİNG.SV kodu

```
module mux4to1_8bit (
   input logic [7:0] in0, in1, in2, in3,
   input logic [1:0] sel,
   output logic [7:0] out
);
   always_comb begin
      case (sel)
         0: out = in0;
         1: out = in1;
         2: out = in2;
         3: out = in3;
         default: out = 0;
      endcase
   end
endmodule
```

```
# KERNEL: --- 8-Bit 4:1 MUX Testi Basliyor ---
# KERNEL: Girisler: in0=aa, in1=bb, in2=cc, in3=dd
# KERNEL: Zaman = 10 | sel = 00 | Cikis = aa (Beklenen: AA)
# KERNEL: Zaman = 20 | sel = 01 | Cikis = bb (Beklenen: BB)
# KERNEL: Zaman = 30 | sel = 10 | Cikis = cc (Beklenen: CC)
# KERNEL: Zaman = 40 | sel = 11 | Cikis = dd (Beklenen: DD)
# KERNEL: --- Test Tamamlandi ---
```

Kodların çıktısı